+3V3 Buck Regulator

VOUT = 0.6*(1+53.6/12)=3.28V

+VBUSF
U1
TLV62568DBV
VIN
SW
EN
FB
GND
L1
4u7
R2
53k6
R3
12k
C1
22u
C4
12p
C5
22u
+3V3
GND

USB Type B

J1
USB Type B
VBUS
D–
D+
SSRX–
SSRX+
SSTX–
SSTX+
GND
GNDD
Shield
SH
GNDU
C25
220p
R1
0R
C2
1u
C3
1u
+VBUSF
+VBUSF
GND
ESD PROTECTION
5V 5.5V 10V
0.15pF
8-CHANNELS
BI-DIR
NC
D1

DP_90R USB.D_N
DP_90R USB.D_P
DP_90R USB.RX_N
DP_90R USB.RX_P
DP_90R USB.TX_N
DP_90R USB.TX_P

USB3 P/N swap OK

C6 1u
C7 1u
C8 1u
C9 1u
C10 22u
C11 1u
C12 1u
C13 1u
C14 1u
C15 1u
C16 1u
+3V3
+2V5
GND

L2
4u7
C17
22u
C18
1u
C19
1u
+1V2
R4
0R
C20 1u
C21 1u
C22 1u
+1V2A

R5
12k
+3V3

VBUS_DET
USB2_DM
USB2_DP
USB3_RXDM
USB3_RXDP
USB3_TXDM
USB3_TXDP
RESET_N/PME_CLEAR
EECS/GPIO0
EEDI/GPIO1
EEDO/LED0/GPIO2
EECLK/LED1/GPIO3
XI
XO
USBRBIAS
TEST
VDDVARIO_1
VDDVARIO_2
VDDVARIO_3
VDD33A
VDD_SW_IN
VDD33_REG_IN
VDD25_REG_OUT
VDD25A_1
VDD25A_2
VDD25A_3
VDD25A_4
VDD12_SW_OUT
VDD12_SW_FB
VDD12CORE_1
VDD12CORE_2
VDD12A_1
VDD12A_2
VDD12A_3
TR0P
TR0N
TR1P
TR1N
TR2P
TR2N
TR3P
TR3N
PME_N/GPIO4
SUSPEND_N/LED2/GPIO5
LED3/GPIO6
PME_MODE/GPIO7
REF_REXT
REF_FILT
EP
IC1
LAN7800_Y9X

C23 100n
C24 100n
R6 12k
+3V3
Using internal OTP
instead of EEPROM

Y1
25MHz
C30 12p
C29 12p
R13 12k
R12 12k

DP_100R MDITR0_P
DP_100R MDITR0_N
DP_100R MDITR1_P
DP_100R MDITR1_N
DP_100R MDITR2_P
DP_100R MDITR2_N
DP_100R MDITR3_P
DP_100R MDITR3_N

LED2
LED3

R10 1k
C28 1u
R11 1k

1G MagJack

J2
HR911130A
MD0+
MD0–
MD1+
MD1–
MD2+
MD2–
MD3+
MD3–
LED_GREEN_A
LED_GREEN_K
LED_YELLOW_A
LED_YELLOW_K
VCC
GND
SHIELD

LED2=Link/Activity
LED3=Duplex/Collision

R7 1k
R8 1k
R9 0R
C27 1u
C26 220p
GNDE

LED2/3 ENABLE –> WRITE 0x00C00000 to 0x10

H1 H2 H3 H4
GND

USB ETHERNET ADAPTER

Sheet: /		
File: USB_Ethernet_Adapter.kicad_sch		
Title:		
Size: A3	Date: 2026-10-03	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1